

An Exact Neumann-Mode Turing Atlas for the Schnakenberg Reaction–Diffusion System

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Abstract

For the Schnakenberg system on a finite interval we derive the unique positive constant equilibrium, its kinetic stability chamber, and the exact two-by-two dispersion matrix for every Neumann mode. Inside the kinetically stable chamber we prove an if-and-only-if continuous wavenumber window, an if-and-only-if finite-domain criterion, an exact strict integer count of unstable modes, and every domain-length entry and exit wall.

1 Frozen model and kinetic chamber

Let $a, b, d_u, d_v, L > 0$ and consider, on $0 < x < L$,

$$\begin{aligned} u_t &= d_u u_{xx} + a - u + u^2 v, \\ v_t &= d_v v_{xx} + b - u^2 v, \end{aligned} \quad u_x = v_x = 0 \quad \text{at } x = 0, L. \quad (1)$$

Put $s = a + b$. A positive spatially homogeneous equilibrium must satisfy $u^2 v = b$ and $a - u + b = 0$. Thus the unique positive spatially homogeneous equilibrium is

$$(u_*, v_*) = (s, b/s^2). \quad (2)$$

The reaction Jacobian there is

$$J = \begin{pmatrix} (b-a)/s & s^2 \\ -2b/s & -s^2 \end{pmatrix}, \quad \tau := \text{tr } J = \frac{b-a}{s} - s^2, \quad \det J = s^2. \quad (3)$$

Thus the kinetic equilibrium is asymptotically stable exactly when $\tau < 0$. At $\tau = 0$ its eigenvalues are $\pm is$; for $\tau > 0$ it has positive spectral abscissa.

Expand perturbations in the normalized Neumann cosine basis. With

$$\mu_n = (n\pi/L)^2, \quad n = 0, 1, 2, \dots, \quad (4)$$

the n th coefficient evolves under

$$M(\mu) = J - \begin{pmatrix} d_u \mu & 0 \\ 0 & d_v \mu \end{pmatrix}. \quad (5)$$

Direct expansion gives

$$T(\mu) = \text{tr } M(\mu) = \tau - (d_u + d_v)\mu, \quad (6)$$

$$D(\mu) = \det M(\mu) = d_u d_v \mu^2 - B\mu + s^2, \quad B := d_v \frac{b-a}{s} - d_u s^2. \quad (7)$$

Consequently the complete modal pair is

$$\lambda_{n,\pm} = \frac{T(\mu_n) \pm \sqrt{T(\mu_n)^2 - 4D(\mu_n)}}{2}. \quad (8)$$

The phrase *kinetic and modal dispersion closure* marks the baseline manuscript: no continuum-wavenumber test has yet been promoted to a finite interval claim.

2 Continuous window and exact finite-domain selection

Define

$$Q = B^2 - 4d_u d_v s^2. \quad (9)$$

Theorem 1 (Complete linear Turing criterion). *Assume $\tau < 0$. A nonempty open interval of unstable positive squared wavenumbers exists if and only if*

$$B > 0, \quad Q > 0. \quad (10)$$

When these inequalities hold, set

$$\mu_{\pm} = \frac{B \pm \sqrt{Q}}{2d_u d_v}. \quad (11)$$

Then $0 < \mu_- < \mu_+$ and the finite Neumann interval has a genuine linear Turing instability if and only if some $n \geq 1$ satisfies

$$\mu_- < \mu_n < \mu_+. \quad (12)$$

Writing $r_{\pm} = (L/\pi) \sqrt{\mu_{\pm}}$, the exact number of unstable modes is

$$N_{\text{unst}} = \max\{0, \lceil r_+ \rceil - \lfloor r_- \rfloor - 1\}. \quad (13)$$

For fixed $n \geq 1$, its precise domain-length chamber is

$$\frac{n\pi}{\sqrt{\mu_+}} < L < \frac{n\pi}{\sqrt{\mu_-}}. \quad (14)$$

Equality at either wall is neutral rather than unstable.

Proof. Because $\tau < 0$, equation (6) is negative for every $\mu \geq 0$. A real two-by-two matrix with negative trace has positive spectral abscissa exactly when its determinant is negative. Indeed, a negative determinant gives two real eigenvalues of opposite signs, whereas a positive determinant and negative trace put both eigenvalues in the open left half-plane.

The polynomial $D(\mu)$ has positive leading and constant coefficients. It is negative somewhere on the positive axis precisely when its vertex is positive and its minimum is negative, which is exactly (10). The two roots are (11); their sum and product are positive, so both are positive and $D < 0$ precisely between them. This proves the continuous criterion and (12) after restricting to the actual Neumann spectrum.

Taking positive square roots of (12) gives $r_- < n < r_+$. The number of positive integers in this strict interval is exactly (13); the ceiling and floor conventions delete integer endpoints. Solving the same strict inequalities for L yields (14). $\square$

At a window endpoint $D = 0$ and $T < 0$, so the two modal eigenvalues are 0 and T : the zero eigenvalue is simple. The phrase *strict interval mode-count closure* marks the first substantive revision: continuous dispersion, discrete mode selection, multiplicity, and length walls now form one if-and-only-if theorem.